

NABEEL REHEMTULLA

CURRICULUM VITAE

nabeelr@u.northwestern.edu
(734) 272-2463

nabeelr.com
github.com/nabeelre

EDUCATION

PhD	Northwestern University Astronomy (Advisor: Adam A. Miller)	<i>In Progress</i>
MS	Northwestern University (GPA: 3.9) Astronomy Integrated Data Science Certificate	2023
BS	University of Michigan - Ann Arbor (GPA: 3.5) Major Astronomy & Astrophysics with Highest Honors Major Interdisciplinary Physics Minor Computer Science	2021

FIRST-AUTHOR PUBLICATIONS

Rehemtulla, et al., "*Pre-training vision models for the classification of alerts from wide-field time-domain surveys*", 2025d, arXiv:2512.11957

Rehemtulla, et al., "*The automation of optical transient discovery and classification in Rubin-era time-domain astronomy*", 2025c, NatAs, 9, 1764R

Li, Chen, **Rehemtulla***, et al., "*StarEmbed: Benchmarking Time Series Foundation Models on Astronomical Observations of Variable Stars*", 2025b, arXiv:2510.06200

*Co-first author and corresponding author

Rehemtulla, et al., "*The BTSbot-nearby Discovery of SN 2024jlf: rapid, autonomous follow-up probes interaction in an 18.5 Mpc Type IIP supernova*", 2025a, ApJ, 985, 241

Rehemtulla, et al., "*The Zwicky Transient Facility Bright Transient Survey. III. BTSbot: Automated Identification and Follow-up of Bright Transients with Deep Learning*", 2024, ApJ, 972, 7

Rehemtulla, et al., "*Non-parametric spherical Jeans mass estimation with B-splines*", 2022, MNRAS, 511, 5536

CO-AUTHOR PUBLICATIONS

Sevilla, et al., "*Multiwavelength Analysis of Six Luminous, Fast Blue Optical Transients*", 2026, arXiv:2601.18926

Barjou-Delayre, et al., "*Study of the anisotropy of cosmic expansion on ZTF type Ia supernovae simulations*", 2026, arXiv:2601.11139

Perley, et al., "*AT2024wpp: An Extremely Luminous Fast Ultraviolet Transient Powered by Accretion onto a Black Hole*", 2026, arXiv:2601.03337

Burgaz, et al., "*ZTF SN Ia DR2 follow-up: Exploring the origin of the Type Ia supernova host galaxy step through Si II velocities*", 2026, A&A, 705, A76

- Kasliwal, et al., "*ZTF25abjmnps (AT2025ulz) and S250818k: A Candidate Superkilonova from a Subthreshold Subsolar Gravitational-wave Trigger*", 2025, ApJ, 995, L59
- Pearson, et al., "*Mid-infrared Dust Evolution and Late-time Circumstellar Medium Interaction in SN 2017eaw*", 2025, ApJ, 993, 213
- Kwok, et al., "*JWST Spectroscopy of SN Ia 2022aaq and 2024gy: Evidence for Enhanced Central Stable Ni Abundance and a Deflagration-to-Detonation Transition*", 2025, arXiv:2510.09760
- Xu, et al., "*AppleCiDER II: SpectraNet -- A Deep Learning Network for Spectroscopic Data*", 2025, arXiv:2510.07215
- Miller, et al., "*The La Silla Schmidt Southern Survey*", 2025, PASP, 137, 094204
- Gupta, et al., "*Transfer learning for transient classification: from simulations to real data and ZTF to LSST*", 2025, MNRAS, 542, L132
- Medina, et al., "*The mass of the Milky Way from outer halo stars measured by DESI DRI*", 2025, arXiv:2508.19351
- Kwok, et al., "*JWST and Ground-based Observations of the Type Iax Supernovae SN 2024pxl and SN 2024vjm: Evidence for Weak Deflagration Explosions*", 2025, ApJ, 989, L33
- Junell, et al., "*Applying multimodal learning to Classify transient Detections Early (AppleCiDER) I: Data set, methods, and infrastructure*", 2025, arXiv:2507.16088
- Yao, et al., "*A Massive Black Hole 0.8 kpc from the Host Nucleus Revealed by the Offset Tidal Disruption Event AT2024tvd*", 2025, ApJ, 985, L48
- Singh, et al., "*Photometry and Spectroscopy of SN 2024pxl: A Luminosity Link Among Type Iax Supernovae*", 2025, arXiv:2505.02943
- Das, et al., "*Low-luminosity Type IIP Supernovae from the Zwicky Transient Facility Census of the Local Universe. I. Luminosity Function, Volumetric Rate*", 2025, PASP, 137, 044203
- Sharma, et al., "*CCSNscore: A Multi-input Deep Learning Tool for Classification of Core-collapse Supernovae Using SED-machine Spectra*", 2025, PASP, 137, 034507
- Sharma, et al., "*Dramatic Rebrightening of the Type-changing Stripped-envelope Supernova SN 2023aew*", 2024, ApJ, 966, 199
- Strotjohann, et al., "*Search for Supernova Progenitor Stars with ZTF and LSST*", 2024, ApJ, 960, 72
- Cooper, et al., "*Overview of the DESI Milky Way Survey*", 2023, ApJ, 947, 37

SELECTED FIRST-AUTHOR CONFERENCE PROCEEDINGS AND ASTRONOTES

AstroNotes are circulars to the transient astronomy community

Rehemtulla, et al., 2025d, "Discovery and classification of SN 2025ahqr, a highly reddened low luminosity Type Iax in M108 (9 Mpc)", TNSAN, 2025-345

Rehemtulla, et al., 2025c, "Swift/UVOT observations of SN 2025qtt 6 minutes after its automatic discovery by BTSbot-nearby", TNSAN, 2025-209

Rehemtulla, et al., 2025b, "Automatic ground- and space-based follow-up of SN 2025qpk following its discovery by BTSbot-nearby", TNSAN, 2025-208

Rehemtulla, et al., 2025a, "Early spectroscopy of SNe 2025ay, 2025mg, and 2025oq autonomously triggered by BTSbot-nearby", TNSAN, 2025-46

Rehemtulla, et al., 2023, "SN 2023tyk: discovery to spectroscopic classification performed fully automatically", TNSAN, 2023-265

♦ *5th most viewed AstroNote of all time*

Rehemtulla, et al., "BTSbot: A Multi-input Convolutional Neural Network to Automate and Expedite Bright Transient Identification for the Zwicky Transient Facility", 2023, Machine Learning for Astrophysics, 27

AWARDS AND FELLOWSHIPS

Presidential Fellowship Northwestern University – the <i>highest honor Northwestern bestows upon a graduate student</i>	2025
Outstanding Student Achievement Award Astroinformatics Conference, Puerto Natales, Chile	2024
2nd prize, Rapid Fire Research 2024 Northwestern University, Department of Physics and Astronomy	2024
1st prize, Visualization Competition (animation) Northwestern University, Computation and Data Exchange (CoDEx)	2024
NASA Illinois Space Grant Consortium Graduate Fellowship (awarded twice)	2023, 2022
Highest Honors in Astronomy & Astrophysics University of Michigan, Department of Astronomy	2021
Chambliss Astronomy Achievement Award American Astronomical Society	2021
Outstanding Undergraduate Poster Michigan Institute for Data Science	2020

TALKS, POSTERS, AND PANELS

Talk, Astroinformatics 2026, Cologne, Germany (upcoming)	February 2026
<i>Invited</i> – Talk, Earth 2.0 Telescope Science Seminar, (remote)	January 2026
Poster, Open SkAI 2025, Chicago, IL	September 2025
<i>Invited</i> – Panelist, Rutgers Summer Transient Soiree, Piscataway, NJ	July 2025
Talk, Rutgers Summer Transient Soiree, Piscataway, NJ	July 2025
<i>Invited</i> – Talk, Foundation Models for Astronomy, New York, NY	May 2025
Talk, Celebrating 20 Years of Swift Discoveries, Florence, Italy	March 2025
<i>Invited</i> – Talk, LS4 Team Meeting, Evanston, IL	March 2025
<i>Invited</i> – Talk, Rutgers University Astronomy Journal Club (remote)	February 2025
Talk, 245 th AAS Meeting, National Harbor, MD	January 2025

Talk, Astroinformatics 2024, Puerto Natales, Chile	December 2024
Poster, Rise_Time 2024, West Lafayette, IN	August 2024
<i>Invited</i> – Talk, A3D3 Multi-messenger Astrophysics Monthly Telecon	November 2023
Poster, International Conference on Machine Learning, Honolulu, HI	July 2023
Talk, The Transient and Variable Universe, Urbana, IL	June 2023
Lecture, ASTRON 111 – Introduction to Astrobiology, Evanston, IL	April 2023
Poster, 241 st AAS Meeting, Seattle, WA	January 2023
Poster, 237 th AAS Meeting <i>DESI Special Session</i> (remote)	January 2021
Talk & Poster, Michigan Institute for Data Science Poster Symposium	November 2020
Poster, Astronomy Undergraduate Poster Symposium, Ann Arbor, MI	April 2018
Not shown: 7 talks and posters at internal DESI and ZTF collaboration events (2021–2025) and 10+ talks, posters, and panels at Northwestern & CIERA internal events (2021–2026)	

TEACHING AND MENTORING

Northwestern University, CIERA

Code/Astro Workshop, Teaching Assistant	Summer 2023
ASTRON 111, Teaching Assistant	Spring 2023
ASTRON 120, Teaching Assistant	Fall 2022
Research Experiences at CIERA for High Schoolers, Project Mentor	Summer 2022
♦ <i>Asher Grossman: Machine learning classification of cosmic explosions</i>	

University of Michigan Department of Astronomy

Academic Success Program, Tutor	Fall 2019
Astronomy Structured Study Group, Co-Facilitator	Fall 2019
♦ <i>Tripled attendance relative to previous years after rehauling curriculum</i>	

Ismaili Center of Detroit (hometown mosque)

Religious Education Center, 7 th & 8 th Grade Teacher	2019 – 2020
English as a Second Language Teacher	Summer 2018

LEADERSHIP

Graduate Student Council Mentorship Committee , NU Dept. of Physics and Astronomy	
Executive Board Member	April 2025 – Present

Ismaili Center of Chicago (local mosque), Chicago, IL

Professionals Network, Midwest - Board Member	September 2023 – Present
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Student Astronomical Society, University of Michigan

Executive Board, Treasurer	2020 – 2021
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Ismaili Center of Detroit (hometown mosque), Southfield, MI

Regional Youth Volunteer Co-Facilitator	2018 – 2021
Youth & Sports Board Regional Representative	2020 – 2021

WORKSHOPS

CCA The Landscape of Astronomical Transients	July 2025
• Theory and Observations of Transients in the Context of LSST	

DESI MWS Team Workshop 2024	August 2024
Coordinating and optimizing use of DESI Milky Way Survey data	
Time-Domain Needles in Rubin's Haystack	April 2024
Time-domain anomaly detection methods for the Rubin Observatory	
Astro Imaging Workshop 2023	July 2023
Image signal processing methods in astronomical data science	
LSST Data Science Fellowship Program Session 16 (<i>Audited</i>)	September 2022
Statistics and Bayesian inference	
Zwicky Transient Facility Summer School	July 2022
Data processing, Bayesian inference, time-series analysis, machine learning	
Code/Astro	June 2022
Best practices for open-source software development in astronomy	

PRESS

33 total media mentions, total reach of 3.1M

Selected articles featuring "*SN 2023tyk: discovery to spectroscopic classification performed fully automatically*" (Rehemtulla et al. 2023)

- ◆ [Northwestern Now](#), "First supernova detected, confirmed, classified and shared by AI"
- ◆ [Space.com](#), "AI just spotted its 1st supernova. Could it replace human explosion hunters?"
- ◆ [Gizmodo](#), "AI Detected a Supernova All on Its Own"
- ◆ [Dropbox Blogs](#), "How an AI-powered telescope is helping astronomers sift through the stars"

Selected articles featuring "*BTSbot: Automated Identification and Follow-up of Bright Transients with Deep Learning*" (Rehemtulla et al. 2024)

- ◆ [Northwestern Now](#), "Sky survey classifies more than 10,000 supernovae"
- ◆ [Caltech News](#), "ZTF Leads to Classification of 10,000 Supernovae"

AWARDED TELESCOPE TIME

2026A

PI, MMT Observatory	3.5 nights
PI, Gemini-N Observatory	10 hours
PI, SOAR Observatory	13 hours
PI, SALT Observatory	3.5 hours
Co-I, Nordic Optical Telescope	20 hours

2025B

PI, MMT Observatory	3.5 nights
PI, Gemini-N Observatory	10 hours
Co-I, Gemini-N/-S Observatory (Large and long program)	>100 hours
Co-I, Nordic Optical Telescope	20 hours
Co-I, W. M. Keck Observatory	1.5 nights

Co-I, Magellan Observatory	1 night
2025A	
PI, MMT Observatory	3 nights
PI, Southern Astrophysical Research (SOAR) Telescope	10 hours
Co-I, Nordic Optical Telescope (two approved programs)	40 hours
Co-I, XMM Newton Space Telescope	~17 hours
Co-I, W. M. Keck Observatory	2 nights
Co-I, Magellan Observatory	1 night
Co-I, SOAR Telescope	10 hours
2024B	
PI, SEDM-KP	104 hours
Co-I, MMT Observatory	4 nights
Co-I, W. M. Keck Observatory	2 nights
2024A	
PI, SEDM-KP	104 hours
Co-I, MMT Observatory	4 nights
Co-I, W. M. Keck Observatory	2.5 nights
Co-I, Very Large Array (<i>Director's Discretionary Time</i>)	~4 hours
2023B	
PI, SEDM-KP	104 hours
Co-I, MMT Observatory	4 nights
Co-I, W. M. Keck Observatory	2.5 nights
2023A	
Co-I, MMT Observatory	4 nights